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Title: Augmented Reality (AR) in Education: Applications and Challenges

By

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Abstract

The paper presents a comprehensive study on the use of augmented reality (AR) technology in science education. The research focuses on the immersion experiences of 180 university students studying in the science education department, with a particular emphasis on factors such as total immersion, engrossment, and engagement. The study utilizes a structured questionnaire to measure the immersion level of students while using AR applications, and the data collection process involves the use of a structural equation model (SEM) to test the hypothesis about the relationships between measured variables. The findings highlight the significance of AR technology in fostering student engagement, creativity, and imagination, and emphasize the need for further exploration and application of AR in educational settings. The document also addresses the potential implications of the study's findings for science education, emphasizing the need for further research with larger sample sizes to validate the model's outcomes and ensure its generalizability to a wider population.

Keywords

Augmented Reality, Science Education, Immersion Experiences, Technology, Usage, Usability, Emotional Investment, Focus of Attention, Structural Equation Model

Section 1: Introduction

Introduction

Augmented Reality (AR) has gained significant attention in the field of education due to its potential to enhance learning experiences and engage students in new and innovative ways. AR technology integrates virtual elements into the real world, creating an interactive and immersive learning environment (Yuen, Yaoyuneyong, & Johnson, 2011). The use of AR in education has been explored in various studies, with a focus on its applications and the challenges associated with its implementation.

AR applications in education have been designed to support learning in different subjects, including geometry, science, and engineering. For example, a study conducted in Mexican schools investigated the use of AR technology to practice basic concepts of geometry, comparing its effectiveness in public and private school settings (Ibáñez, et al., 2019). The findings of this study provide insights into the impact of AR on student learning outcomes and motivation in diverse socio-economic contexts.

In addition to geometry, AR has been utilized to teach subjects where real-world experiences are limited, such as astronomy and geography (Shelton & Hedley, 2002). Furthermore, AR applications have been developed to facilitate learning in science and technology education, with a focus on collaborative learning and differentiated instruction (Salar, Arici, Caliklar, & Yilmaz, 2019). These applications aim to enhance student engagement, creativity, and control over their learning process.

The potential of AR in education extends beyond traditional classroom settings, as it offers opportunities for students to interact with virtual objects and environments. For instance, AR technology has been used to create authentic learning environments suitable for various learning styles, fostering student creativity and imagination (Klopfer & Yoon, 2004). Moreover, AR applications have been explored in fields such as medical training simulations, engineering, e-commerce, and architecture, demonstrating the diverse range of educational possibilities (Liu, Jenkins, Sanderson, Fabian, & Russell, 2010; Zhu, Owen, Li, & Lee, 2004).

Despite the promising applications of AR in education, there are challenges associated with its implementation. Research indicates the need for larger sample studies to understand the long-term effects of AR technology on student learning and motivation (Salar, Arici, Caliklar, & Yilmaz, 2019). Additionally, the development of AR applications requires consideration of tracking, display, and input technologies, as well as calibration and registration issues (Zhou, Dun, & Billinghurst, 2008). These technical challenges highlight the complexity of integrating AR into educational settings.

In conclusion, the use of AR technology in education presents both opportunities and challenges. While AR has the potential to enhance student engagement, motivation, and learning outcomes, there is a need for further research to explore the effectiveness of AR applications in different educational contexts and to address the challenges associated with the implementation of AR technology in the classroom (Yuen, Yaoyuneyong, & Johnson, 2011). As the field of AR in education continues to evolve, it is essential to consider the ethical and practical implications of integrating AR technology into the learning environment (Arici, Salar, Caliklar, & Yilmaz, 2019).

Section 2: Literature Review

Augmented Reality (AR) and Virtual Reality (VR)

Augmented Reality (AR) enhances real-world interactions by overlaying digital information that users can interact with, significantly improving learning experiences and student motivation by allowing digital and real-world objects to coexist and interact seamlessly (Ibáñez et al., 2019). Conversely, Virtual Reality (VR) creates a completely immersive digital environment that substitutes the real world, typically accessed through devices like head-mounted or hand-held controllers. This immersion makes users feel directly engaged with the virtual environment, enhancing educational outcomes in fields such as medicine, architecture, and history by allowing students to experience places and situations they cannot physically visit, thereby facilitating deeper engagement and learning (Balog & Pribeanu, 2010).

Differences and Interconnections

Differences and interconnections between the use of Augmented Reality (AR) in education can be observed in various aspects. One key difference lies in the socio-economic contexts in which AR technology is implemented. For instance, a study conducted in Mexico compared the use of AR technology in public and private schools, revealing varying impacts on student motivation and learning outcomes based on the type of school. Additionally, the design of AR applications for educational purposes differs based on the official guidelines set by educational programs, with the aim of allowing students to practice basic concepts of Geometry (Ibáñez Marí et al., 2019). Furthermore, the research on AR applications within education has explored various fields and disciplines, such as medical training simulations, engineering, and e-commerce, indicating the diverse applications and potential benefits of AR technology in different educational contexts (Liu et al., 2010; Liarokapis et al., 2004; Zhu et al., 2004). Moreover, the use of AR technology in education has been found to foster student creativity, enhance collaboration, and create authentic learning environments suitable for various learning styles (Klopfer & Yoon, 2004). Finally, the future of AR in education is expected to continue evolving, with the widespread use of AR in education predicted within the next 2 to 3 years across college campuses in the United States (Dede, 2008; Hamilton, 2011).

History of AR

The figure - History of AR - a brief timeline in the paper "Augmented Reality: An Overview and Five Directions for AR in Education" (Yuen, S., Yaoyuneyong, G., & Johnson, E., 2011), outlines the key milestones and developments in the field of AR, showcasing the evolution of this technology over time.

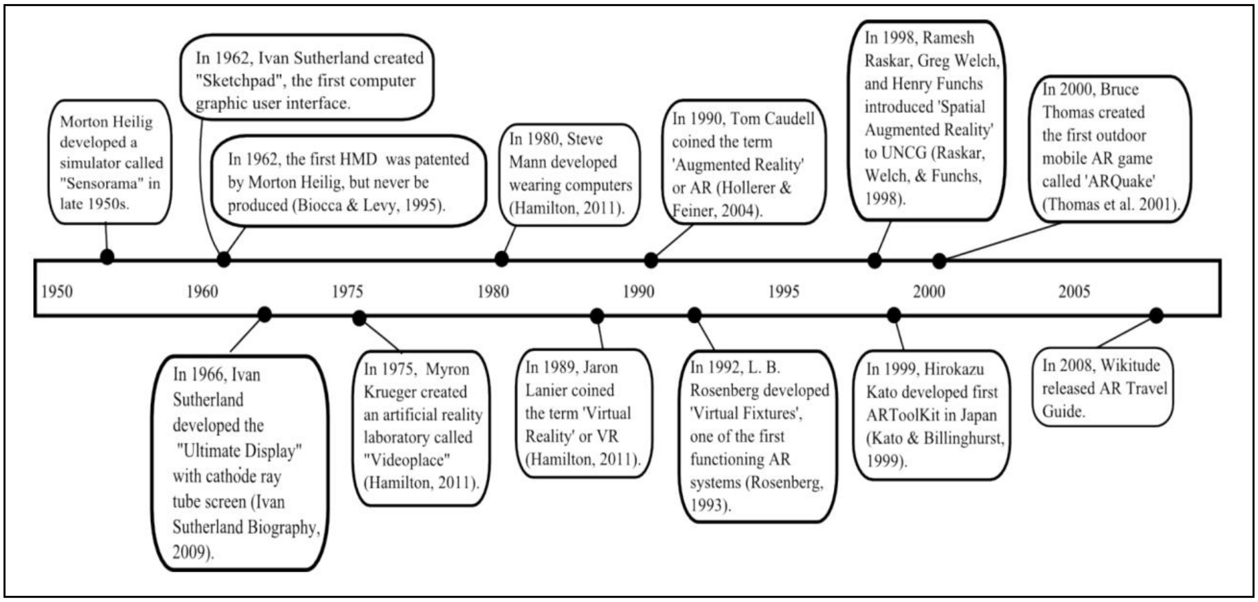

Figure 1: History of AR - a brief timeline

Augmented Reality: An Overview and Five Directions for AR in Education (Yuen, S., Yaoyuneyong, G., & Johnson, E., 2011)

The history of Augmented Reality (AR) can be traced back to the 1960s when Ivan Sutherland developed the first head-mounted display system, known as the “Sword of Damocles” (Ivan Sutherland Biography | World of Computer Science Biography). This marked the beginning of AR technology, which has since evolved to become more accessible to the masses, especially with the implementation of Flash-based AR detection algorithms and the growing popularity of mobile platforms such as iOS and Android. Over the years, AR has seen significant developments, with the emergence of marker tracking and head-mounted display calibration for video-based AR conferencing systems (Kato & Billinghurst, 1999). The timeline of AR's history is a testament to the continuous advancements and innovations that have shaped its evolution into a valuable educational tool.

Furthermore, the history of AR in education has been characterized by the exploration of its potential to revolutionize the learning experience. Educators have envisioned AR as a means to achieve ubiquitous learning, providing immediate access to location-specific information compiled from various sources. The impact of AR on society has been recognized, with predictions that AR will see widespread use on college campuses within the next 2 to 3 years. This highlights the growing significance of AR in education and its potential to positively impact learning, creative inquiry, and education (NMC, 2011). The history of AR reflects its journey from a novel technology to a promising tool for enhancing the educational experience.

Historical Development of AR in Education

The historical development of Augmented Reality (AR) in education has been a result of extensive research and technological advancements. Over the years, researchers have explored the use of AR applications in various fields and disciplines, many of which are directly or indirectly related to education (Liu, Jenkins, Sanderson, Fabian, & Russell, 2010). For instance, AR has been utilized in medical training simulations, engineering concepts exploration, and even in the field of e-commerce and architecture (Sielhorst, Obst, Burgkart, Riener, & Navab, 2004; Liarokapis et al., 2004; Zhu, Owen, Li, & Lee, 2004). The development of AR technology has opened up new possibilities for creating authentic learning environments suitable for various learning styles, fostering student creativity and imagination, and enhancing collaboration between students and instructors (Klopfer & Yoon, 2004; Billinghurst, 2002).

Furthermore, the historical development of AR in education has been driven by the increasing focus on incorporating AR elements within mobile applications, leading to a significant rise in the number of AR capable apps available (Juniper Research, 2011). This surge in AR development has been particularly prominent in the areas of marketing and entertainment, with a growing emphasis on small mobile devices such as Android Phones and iPhones (Hamilton, 2011). Additionally, the emergence of AR technology has paved the way for the transformation of school field trips, replacing traditional paper question sheets with just-in-time information access through learners' smartphones' GPS (Thomas et al., 2001). These advancements have significantly contributed to the historical evolution of AR in education, shaping its applications and challenges in the modern educational landscape.

Applications of AR in Education

Augmented Reality (AR) has numerous applications in the field of education, offering innovative ways to enhance the learning experience. AR can be used to provide immersive experiences that allow students to interact with virtual objects in real-world environments, making abstract concepts more tangible and easier to understand (Akçayır & Akçayır, 2017). Additionally, AR can be utilized to create engaging and interactive learning materials that cater to different learning styles, fostering student creativity and imagination (Klopfer & Yoon, 2004). Furthermore, AR applications can help students take control of their learning at their own pace and on their own path, providing a personalized learning experience (Hamilton & Olenewa, 2010).

One of the key applications of AR in education is the ability to provide students with real-world experiences that would otherwise be unfeasible, such as in subjects like astronomy and geography (Shelton & Hedley, 2002). AR can also enhance collaboration between students and instructors, as well as among students, by creating an authentic learning environment suitable to various learning styles. Moreover, AR can be used to facilitate learning in various disciplines, such as mathematics, geometry, engineering, and science, by creating interactive and immersive learning experiences (Kaufmann, 2003; Liarokapis et al., 2004).

In the context of medical education, AR has been explored for applications in general anesthesia, medical training simulations, and medical display literature, offering potential benefits for hands-on learning experiences in the medical field (Liu et al., 2010; Sielhorst et al., 2004; Sielhorst et al., 2008). Additionally, AR can be used to support the teaching of architecture, interior design, and e-commerce, providing students with practical and visual learning opportunities (Billinghurst & Henrysson, 2009; Phan & Choo, 2010; Zhu, Owen, Li, & Lee, 2004).

Furthermore, AR applications in education extend to the development of AR books and educational gaming experiences, offering new interfaces for storytelling and interactive learning. AR gaming can be utilized to assist students in grasping class concepts and can be applied to a range of disciplines, including archaeology, history, anthropology, and geography, providing educators with powerful new ways to demonstrate relationships and connections (Yuen, Yaoyuneyong, & Johnson, 2011).

Impact of AR on Student Learning

The impact of Augmented Reality (AR) on student learning has been a subject of extensive research, with findings indicating that the use of AR applications in educational settings can significantly influence student outcomes. Studies have shown that AR technology has the potential to engage, stimulate, and motivate students to explore class material, leading to improved learning outcomes (Contero, 2017; Chang, & Hwang, 2018). Additionally, AR has been found to enhance collaboration between students and instructors, foster student creativity and imagination, and create an authentic learning environment suitable for various learning styles (Billinghurst, 2002; Klopfer & Squire, 2008). Furthermore, the use of AR in education has been recognized as a valuable tool for helping students achieve real-world goals through interaction with objects from virtual environments, thereby contributing to the development of creative skills (Bower, 2014).

Moreover, a case study conducted in Mexican public and private schools revealed that AR technology had a positive impact on learning-related outcomes of middle-school students, with students using AR-based learning environments scoring higher in post-tests compared to those using web-based applications. The study also found that the augmented reality learning environment was more effective in promoting learning outcomes in public schools compared to private schools, highlighting the potential of AR to be an effective learning tool in different socio-economic contexts (Ibáñez

Marí.Blanca., Portillo A.U., Cabada Ramó.Zatarain. & Barrón Marí.Lucí., 2019). These findings underscore the significance of AR in education and its potential to positively impact student learning outcomes across diverse educational settings.

Pedagogical Approaches for AR Integration

Augmented Reality (AR) in education offers various pedagogical approaches for integration into the classroom. One approach is to use AR to enhance collaboration between students and instructors, fostering student creativity and imagination (Klopfer & Yoon, 2004). This approach allows students to take control of their learning at their own pace and on their own path, creating an authentic learning environment suitable to various learning styles (Classroom Learning with AR, 2010). Additionally, AR can be utilized to teach subjects where students could not feasibly gain real-world first-hand experience, such as astronomy and geography (Shelton & Hedley, 2002). By integrating AR into the curriculum, educators can provide students with immersive learning experiences that go beyond traditional classroom settings, enhancing their understanding of complex concepts (Liu, Jenkins, Sanderson, Fabian, & Russell, 2010).

Another pedagogical approach for AR integration in education is to utilize AR applications to develop students' problem-solving skills and consolidate science concepts (Guzey and Roehrig 2009; Slykhuis and Krall 2011). This approach aligns with the research indicating that the use of AR technology in education provides benefits through the acquisition of knowledge and experience that can be abstract, difficult to understand, hard to observe, and possibly hazardous in a class environment (Bower et al., 2014). Furthermore, AR can be used to engage, stimulate, and motivate students to explore class material, leading to better performance outcomes and increased motivation toward class (Billinghurst 2002; Dalgarno and Lee 2010; Kye and Kim 2008; Lee et al. 2010). By incorporating AR into the learning environment, educators can create dynamic and interactive experiences that cater to the diverse needs of students, ultimately enhancing their academic achievement and motivation (Ibáñez et al., 2015).

Section 3: Factors Influencing Immersion and Engagement in AR Education

Flow and Immersion

Flow and immersion are essential components of the augmented reality (AR) experience in education. The use of AR technology has been shown to increase students' motivation in the lesson, create a more complete focus through an immersion experience, and elicit positive emotional responses (Cai et al., 2013; Bressler & Bodzin, 2013, 2016; Huang et al., 2016). The flow and immersion experience of students while using AR is a key focus of research in this area (Kye and Kim, 2008). Additionally, the relationship between interest, usability, emotional investment, focus of attention, presence, and flow has been examined for university students studying in the science education department who used AR technology, with a model developed to explain 71% of the variance in university students' flow experiences (Salar et al., 2020). Furthermore, the impact of AR technology on learning outcomes and motivation of students from public and private Mexican schools has been investigated, showing that AR technology can be exploited as an effective learning environment for helping middle-school students practice the basic principles of Geometry (Ibáñez et al., 2019).

Measuring Immersion

Measuring immersion in the context of augmented reality (AR) technology in education is a crucial aspect that requires careful consideration. The study conducted by Riza Salar, Faruk Arici, Seyma Caliklar, and Rabia M. Yilmaz (2020) delves into the immersion experiences of university students studying in the science education department. The research design adopted a correlational method and employed a structural equation model to analyze the relationship between interest, usability, emotional investment, focus of attention, presence, and flow. The study revealed that the model developed was capable of explaining 71% of the variance in university students' flow experiences, with a significant influence of focus of attention and presence on their flow experiences. Additionally, the study highlighted the influence of emotional investment and presence of university students on their focus of attention, as well as the impact of usability and emotional investment on their interest (Salar et al., 2020).

Furthermore, the research by Ferrer-Torregrosa et al. (2015) emphasizes the educational benefits of AR technology, including increasing students' motivation in the lesson, presenting an effective and efficient learning environment, and eliciting positive emotional responses. The study also highlights the role of AR in enabling students to experience flow and creating a more complete focus through an immersion experience resulting from interesting activities (Ferrer-Torregrosa et al., 2015). This underscores the significance of measuring immersion in AR applications to enhance the learning experiences of students and promote their engagement and interest in educational settings. Additionally, the study by Ibáñez et al. (2019) provides insights into the impact of AR technology on academic achievement and motivation of students from public and private Mexican schools, offering valuable perspectives on the practical implications and limitations of AR/VR applications in education (Ibáñez et al., 2019).

Interest

Interest in augmented reality (AR) technology in education has been a subject of extensive research, with a focus on its potential to engage and motivate students. The use of AR applications has been shown to positively affect interest in various educational settings, as evidenced by studies conducted in both public and private Mexican schools (Ibáñez et al., 2015). The literature on AR applications emphasizes the importance of increasing student interest through the development of different teaching methods and materials to improve interaction in the classroom (Fonseca et al., 2014). Furthermore, the research indicates that students' use of technology in daily life and in an educational context is already at a high level, which suggests that the use of AR applications should be encouraged in all classes (Salar et al., 2020).

Moreover, the impact of AR on student interest has been explored from various perspectives, including personal characteristics, learning environment related to personal interest, and situational interest (Renninger, Hidi, and Krapp, 2014). The literature also presents a four-stage model of interest, which explains the concept in terms of situational interest, encouraged situational interest, emerging individual interest, and improved personal interest (Hidi and Renninger, 2006). This model suggests that students are able to sustain their internal work motivation through prolonged interest in the subject, which in turn boosts their learning achievement (Ainley, 2007). Additionally, the use of AR in education has been recognized as a valuable tool for helping students achieve real-world goals through interaction with objects from virtual environments, thereby contributing to the development of creative skills (Bower, 2014). These findings underscore the significance of AR technology in fostering student interest and engagement in educational settings (Salar et al., 2020).

Focus of Attention

Augmented Reality (AR) in education has brought about a shift in the focus of attention for students. According to the research conducted by Ibáñez, Portillo, Cabada, and Barrón (2019), the use of AR technology has led to an increased focus of attention among students in middle-school Geometry courses. The immersive nature of AR applications has captivated the attention of students, leading to a more engaged and interactive learning experience. Additionally, the study by Salar, Arici, Caliklar, and Yilmaz (2020) highlights that the focus of attention is a crucial aspect of AR immersion experiences for university students studying in the science education department. The relationship between the focus of attention and the presence of university students has been found to influence their flow experiences, indicating the significance of attention in AR-based learning environments.

Furthermore, the study by Yuen, Yaoyuneyong, and Johnson (2011) emphasizes that AR applications in education have the potential to enhance the focus of attention among students. The interactive and visually stimulating nature of AR technology captures the attention of students, leading to increased engagement and focus during learning activities. Moreover, the research conducted by Contero (2017) and Chang and Hwang (2018) suggests that the use of AR in educational settings promotes a heightened focus of attention among students, contributing to a more immersive and effective learning experience. Overall, the integration of AR technology in education has demonstrated its ability to positively impact the focus of attention among students, leading to enhanced learning outcomes and motivation (Ibáñez et al., 2019).

Flow

Flow, in the context of augmented reality (AR) in education, refers to the state of complete immersion and engagement experienced by students while using AR technology. This state of flow is characterized by a deep focus, heightened motivation, and positive emotional responses (Huang et al., 2016). The literature on AR applications in education suggests that AR can elicit a sense of flow and immersion in students, leading to increased motivation and a more complete focus on the learning activities (Cheng et al., 2015). Additionally, the use of AR technology has been found to facilitate the experience of flow and immersion, which can contribute to a more effective and efficient learning environment (Iordache et al., 2012). Furthermore, the flow and immersion experience of students while using AR is a key focus of research in this area (Bressler and Bodzin, 2013, 2016). Overall, the application of AR in education has the potential to create a conducive environment for students to experience flow, leading to enhanced learning outcomes and motivation (Cai et al., 2013; Ferrer-Torregrosa et al., 2015; Sumadio and Rambli 2010; Wojciechowski and Cellary 2013).

Section 4: Hypotheses

There are 6 hypotheses placed in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" (Salar et al., 2020), which are as follows:

H1. Usability positively affects university students' interest

H2. Emotional investment positively affects university students' interest

H3. Focus of attention and presence of university students have an influence on their flow experiences

H4. Emotional investment and presence of university students influences their focus of attention

H5. Usability and emotional investment of the students influences their interest

H6. Students' attitude towards use of digital learning environments positively affects their behavioral intention to use AR/VR applications for their studies

Hypothesis 1 "Usability positively affects university students' interest " states that usability has a positive effect on university students' interest. This means that the ease of use and functionality of AR technology will influence students' level of interest in using it for educational purposes.

Hypothesis 2 "Emotional investment positively affects university students' interest " suggests that emotional investment positively affects university students' interest. This means that the emotional connection and investment students have in the AR technology will impact their level of interest in using it for their studies.

Hypothesis 3 “Focus of attention and presence of university students have an influence on their flow experiences” states that the focus of attention and presence of university students have an influence on their flow experiences. This means that the level of focus and presence students have while using AR technology will affect their overall immersive experience and flow.

Hypothesis 4 “Emotional investment and presence of university students influences their focus of attention” suggests that emotional investment and presence of university students influence their focus of attention. This means that the emotional investment and sense of presence students have while using AR technology will impact their level of focus and attention.

Hypothesis 5 “Usability and emotional investment of the students influences their interest” states that usability and emotional investment of the students influence their interest. This means that the ease of use and emotional connection students have with AR technology will influence their level of interest in using it for their studies.

Hypothesis 6 “Students’ attitude towards use of digital learning environments positively affects their behavioral intention to use AR/VR applications for their studies“ suggests that students’ attitude towards the use of digital learning environments positively affects their behavioral intention to use AR/VR applications for their studies. This means that students' overall attitude towards digital learning environments will influence their intention to use AR/VR applications for educational purposes.

Section 5: Method and Participation

The research in "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, used a correlational design and structural equation modeling (SEM) to analyze AR immersion experiences among 180 university science students (32 males, 148 females), explaining 71% of the variance in their flow experiences. This analysis provided insights into how interest, usability, emotional investment, focus of attention, presence, and flow correlate within educational AR settings. The findings underscore the potential of AR in enhancing learning, with calls for further studies using larger samples to validate results and deeper qualitative research to explore these variables comprehensively, thereby advancing understanding of AR's educational impact (Salar et al., 2020).

Section 6: Data Collection Tools

The data collection tools used in the study "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, included the ARI questionnaire developed by Georgiou and Kyza (2017) to measure the immersion level of students while using AR applications. This questionnaire utilized a 7-point Likert-type scale ranging from "totally disagree" (1) to "totally agree" (7) and consisted of 21 items, 3 dimensions, and 6 sub-factors. The dimensions of the scale were "total immersion," "engrossment," and "engagement." Sub-factors included "flow," "presence," "emotional investment," "focus of attention," "usability," and "interest" (Fonseca, Marti, Redondo, Navarro, & Sanchez, 2014). The use of this questionnaire allowed for a comprehensive assessment of the students' experiences and interactions with AR technology, providing valuable insights into their immersion experiences and the effectiveness of AR applications in educational settings.

	f	%
Gender		
Male	32	17.8
Female	148	82.2
Age	M	
Mean age	20	
Minimum age	17	
Maximum age	28	
Students' level	*f*	%
Freshman	28	15.6
Sophomore	48	26.7
Junior	55	30.6
Senior	49	27.2

Table 1: Demographic characteristics of sample group

The Table 1: Demographic characteristics of sample group in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, provides a comprehensive overview of the participants involved in the study. It includes details such as the number of male and female students, their prior experience with AR, and the technology use by the sample group. The table also presents information on the personal IT devices used by the students, their internet connection methods, and their demographic characteristics. The data collected from the 180 university students studying in the science education department, as shown in Table 1, offers valuable insights into the sample group's background and technological exposure (Salar et al., 2020).

	f	%		*f*	%
Personal devices used			Stated purposes of Internet usage in daily life		
Desktop computer	35	19.4	Researching	171	95.0
Laptop	94	52.2	Playing games	102	56.7
Tablet	83	46.1	Spending time on social networks	152	84.4
Smartphone	179	99.4	Meeting new people	25	19.9
Other	2	1.1	Online shopping	129	71.7
Internet connection methods			Following news	141	78.3
Mobile internet (3G/4G/4.5G)	169	93.9	Spending time on video sharing websites	108	60.0
Wi-Fi	152	84.4	Chatting	69	38.3
Wired Internet connection	18	10.0	Using e-mail	82	45.6
Prior experience with AR			Doing homework	162	90.0
I do not know anything about AR	137	76.1	Listening to music	170	94.4
I have limited information about AR	43	23.9	Sharing files	107	59.4
IT use skill level in daily life			IT skill level for educational purposes		
1 point	2	1.11	1 point	2	1.11
2 points	13	7.22	2 points	18	10
3 points	67	37.22	3 points	69	38.33
4 points	73	40.55	4 points	69	38.33
5 points	25	13.88	5 points	22	12.22
Mean	3.59		Mean	3.51	

Table 2: Summary of technology use by sample group.

The Table 2: Summary of technology use by sample group in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, provides valuable insights into the technology usage patterns of the participants. The data reveals that the majority of the students used personal IT devices such as laptops (94%) and smartphones (99.4%) for their daily activities. Additionally, about 84.4% of the students used Wi-Fi as their primary internet connection method, indicating a high level of connectivity and access to online resources. Furthermore, the fact that 76% of the students had not previously heard of AR technology highlights the potential novelty effect of short-term involvement with this new technology, which should be considered in future studies (Salar et al., 2020). This information underscores the importance of understanding the technological background and familiarity of participants when implementing AR applications in educational settings.

Georgiou and Kyza's (2017) studies were conducted in Greek and their articles published in English. In this study, as the participants were Turkish, the scale items had to be translated into Turkish by the authors. Three experts checked the Turkish and English items on the scale. English and Turkish language experts were also consulted in order to ensure, as far as possible, the validity of the measures. To ensure reliability, the regression weights of the items were calculated first. One item from the usability factors was excluded due to low regression weight. Cronbach's alpha value of the scale was calculated as 0.913 for this study. Each of the factors scored high for reliability in this study: "interest" ($\alpha = 0.89$), "usability" ($\alpha = 0.69$), "emotional investment" ($\alpha = 0.78$), "focus of attention" ($\alpha = 0.80$), "presence" ($\alpha = 0.87$), and "flow" ($\alpha = 0.79$). In this light, the measurement tool employed can be considered reliable.

Section 7: Data Collection Process

The data collection process in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, involved the use of a structured questionnaire developed by Georgiou and Kyza (2017) to measure the immersion level of students while using AR applications. This questionnaire utilized a 7-point Likert-type scale with 21 items, 3 dimensions, and 6 sub-factors, including "total immersion," "engrossment," and "engagement." The dimensions of the scale were further broken down into sub-factors such as "flow," "presence," "emotional investment," "focus of attention," "usability," and "interest". The sample group for the study consisted of 180 university students studying in the science education department, with 32 males and 148 females, who were training to become science teachers in secondary school. The students were voluntary participants, and the data collection process involved the use of a structural equation model (SEM) to test the hypothesis about the relationships between measured variables in the given population (Salar, Arici, Caliklar, & Yilmaz, 2020).

Furthermore, the study also gathered details about the students' use of personal IT devices, Internet connection methods, and the purposes for which they used the Internet in their daily lives. Information technology (IT) skill levels, both in daily life and in the educational context, were also recorded as part of the data collection process. The ARI questionnaire, which was employed to measure the immersion level of students while using AR applications, was designed to capture the students' experiences and perceptions of AR technology, including their emotional investment, focus of attention, and interest.

The study aimed to investigate the relationship between interest, usability, emotional investment, focus of attention, presence, and flow for university students studying in the science education department who used AR technology (Salar et al., 2020).

Section 8: Data Analysis

Data analysis of the study "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, revealed that the research design adopted was a correlational method, which is an established experimental research method. The study included 180 university students studying in the science education department, with 32 males and 148 females, and the data obtained were analyzed according to the structural equation model (SEM). The study developed a model capable of explaining 71% of the variance in university students' flow experiences, indicating the significant impact of AR technology on their immersion experiences. Additionally, the study showed that emotional investment and presence of university students influenced their focus of attention, and usability and emotional investment of the students influenced their interest (Salar et al., 2020). Further adding to the complexity of the system, there are public and private schools, the former are publicly subsidized whereas the latter derive their resources from student fees (Santiago, McGregor, Nusche, Ravela, & Toledo, 2012).

	Interest	Usability	Emotional investment	Focus of attention	Presence	Flow
Interest	1					
Usability	0.343^{**}	1				
Emotional investment	0.653^{**}	0.197^{**}	1			
Focus of attention	0.622^{**}	0.133	0.728^{**}	1		
Presence	0.433^{**}	0.111	0.544^{**}	0.681^{**}	1	
Flow	0.453^{**}	0.140	0.556^{**}	0.696^{**}	0.647^{**}	1

$^{**}p < .01$

Table 3: Relationship levels between variables.

The Table 3: Relationship levels between variables" in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, presents the relationship levels between interest, usability, emotional investment, focus of attention, presence, and flow for university students using AR technology. The table provides a comprehensive overview of the interconnections between these key variables, shedding light on the complex dynamics at play in the context of augmented reality immersion experiences for science education students. The findings from this table highlight the intricate web of relationships between different factors, emphasizing the need for a holistic understanding of the various elements influencing students' experiences with AR technology in the educational setting (Salar et al., 2020). This analysis contributes to a deeper comprehension of the factors that contribute to the overall immersion experiences of university students studying in the science education department.

Section 9: Findings

The findings of the study "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, revealed that the relationship between interest, usability, emotional investment, focus of attention, presence, and flow significantly impacts university students' experiences with augmented reality (AR) technology. The research design, a correlational method, demonstrated that the focus of attention and presence of university students have a direct influence on their flow experiences, while emotional investment and presence influence their focus of attention. Additionally, the study highlighted that the usability and emotional investment of the students significantly influence their interest in AR technology. Further research is recommended to explore these relationships in more depth and with larger sample sizes to validate the findings (Salar et al., 2020).

Factors	M	sd
Interest	6.28	0.831
Usability	5.93	1.134
Emotional investment	5.69	1.134
Focus of attention	5.51	1.282
Presence	5.11	1.415
Flow	5.20	1.481

Table 4: Descriptive data for each factor

The Table 4: Descriptive data for each factor in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, provides descriptive data for each factor. The table includes information on the mean, standard deviation, skewness, and kurtosis for factors such as total immersion, engrossment, and engagement. It also presents the descriptive statistics for the sub-factors including flow, presence, emotional investment, focus of attention, usability, and interest. This data is essential for understanding the distribution and variability of the factors and sub-factors being analyzed in the study (Salar, Arici, Caliklar, & Yilmaz, 2020).

Section 10: Discussions

The discussion section of the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, delves into the implications of the study's findings. The authors highlight the need for further research with larger sample sizes to validate the model's outcomes and ensure its generalizability to a wider population. Additionally, they emphasize the importance of considering the novelty effect of short-term involvement with AR technology, suggesting that future studies should explore the long-term impact of AR immersion experiences on students' learning and engagement. The discussion also underscores the influence of emotional investment and presence on students' focus of attention, shedding light on the interconnected nature of these variables within the AR immersion context (Salar et al., 2020). Furthermore, the authors advocate for qualitative research to delve deeper into the correlations between the variables used in the model, emphasizing the need for a comprehensive understanding of the underlying mechanisms driving students' flow experiences in AR immersion.

The discussion section also addresses the potential implications of the study's findings for science education. It underscores the significance of AR technology in fostering student engagement, creativity, and imagination, aligning with the broader educational benefits of AR applications (Klopfer & Yoon, 2004). Moreover, the discussion emphasizes the need for educators to consider the diverse learning styles of students and the potential of AR to create an authentic learning environment suitable for accommodating these differences (Wu et al., 2010). The authors also stress the importance of leveraging AR to enhance collaboration between students and instructors, highlighting its potential to facilitate interactive and immersive learning experiences in science education (Billinghurst, 2002). Overall, the discussion section provides valuable insights into the implications of the study's findings for the integration of AR technology in science education, emphasizing the need for further exploration and application of AR in educational settings (Salar et al., 2020).

Section 11: Tested Model

The tested model in the study provides a comprehensive understanding of the factors influencing university students' AR immersion experiences, shedding light on the interplay between various elements such as interest, usability, emotional investment, focus of attention, presence, and flow. This model is a crucial contribution to the field of science education, as it offers insights into the dynamics of AR technology usage among university students and its impact on their learning experiences. By analyzing the relationships between different variables, the model provides a framework for understanding the complexities of AR immersion experiences and their implications for science education (Salar et al., 2020).

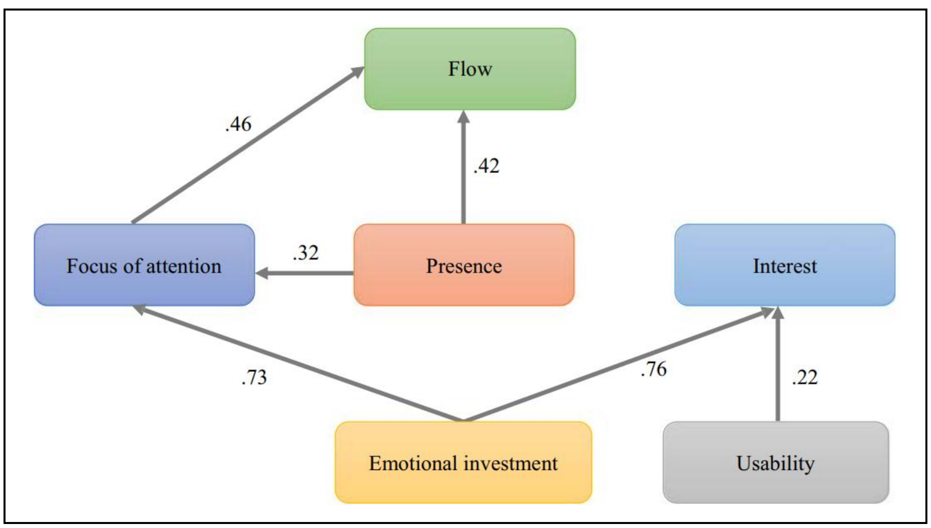

Figure 2: Tested Model

"A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" (Salar, Arici, Caliklar, & Yilmaz, 2020)

The figure Tested - model in this study is a significant aspect of the research conducted in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" (Salar et al., 2020). The model aims to explain the augmented reality (AR) immersion experiences of university students studying in the science education department. It demonstrates the relationship between interest, usability, emotional investment, focus of attention, presence, and flow for students using AR technology. The model was developed based on a correlational method and structural equation model analysis, capable of explaining 71% of the variance in university students' flow experiences. It indicates that the focus of attention and presence of university students have an influence on their flow experiences, and emotional investment and presence of university students influence their focus of attention. Additionally, usability and emotional investment of the students influence their interest, providing valuable insights into the factors affecting students' AR immersion experiences (Salar et al., 2020).

Section 12: Results

The results of the study on "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, revealed a significant relationship between interest, usability, emotional investment, focus of attention, presence, and flow experiences. The research design adopted a correlational method and analyzed data using a structural equation model, which led to the development of a model capable of explaining 71% of the variance in university students' flow experiences. The study found that the focus of attention and presence of university students have a direct influence on their flow experiences, indicating the importance of these factors in shaping the overall AR immersion experience. Additionally, the study demonstrated that emotional investment and presence of university students influence their focus of attention, highlighting the interconnected nature of these variables within the AR learning environment. Furthermore, the results indicated that usability and emotional investment of the students have a direct impact on their interest, emphasizing the role of these factors in shaping students' engagement with AR technology in the context of science education (Salar et al., 2020).

The findings of the study provide valuable insights into the factors that contribute to the effectiveness of AR technology in enhancing the learning experiences of university students studying in the science education department. The results underscore the significance of emotional investment, usability, focus of attention, and presence in shaping students' engagement and interest in AR applications, highlighting the multifaceted nature of the AR immersion experience. Moreover, the development of a model capable of explaining a substantial portion of the variance in university students' flow experiences represents a significant contribution to the understanding of AR technology's impact on student learning and engagement in the context of science education (Salar et al., 2020). The study's results lay the groundwork for future research and the development of targeted interventions aimed at optimizing the use of AR technology in science education, ultimately contributing to the advancement of immersive and effective learning experiences for university students.

Hypothesis	Methods	Standardized regression weights	*p*	Results
H_1	Usability → interest	0.223	0.003	Accepted
H_2	Emotional investment → interest	0.755	0.000	Accepted
H_3	Presence → focus of attention	0.321	0.000	Accepted
H_4	Emotional investment → focus of attention	0.725	0.000	Accepted
H_5	Focus of attention → flow	0.461	0.000	Accepted
H_6	Presence → flow	0.424	0.003	Accepted

Table 5: Tested hypothesis results

The Table - Tested hypothesis results in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, presents the findings of the empirical study conducted to test the research model. The results indicate that the hypotheses related to the relationship between interest, usability, emotional investment, focus of attention, presence, and flow were tested and validated. The study found that the focus of attention and presence of university students have a significant influence on their flow experiences, supporting the proposed model. Additionally, the results showed that emotional investment and presence of university students influence their focus of attention, further validating the model's constructs. Moreover, the study revealed that usability and emotional investment of the students significantly influence their interest, providing valuable insights into the factors affecting students' engagement with augmented reality technology in science education (Salar et al., 2020).

Goodness of fit	Perfect	Acceptable values	Model values	Fit
$\chi 2/df$	< 2	2–5	1.488	Perfect
RMSEA	≤ 0.05	≤ 0.08	0.052	Perfect
AGFI	≥ 0.95	≥ 0.85	0.846	Acceptable
NFI	≥ 0.95	≥ 0.90	0.896	Acceptable

Table 6: Goodness of fit value range and model value

The Table 6: Goodness of fit value range and model value in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, provides important insights into the model's performance. The table presents the goodness of fit values and model values, which are crucial for evaluating the accuracy and reliability of the structural equation model used in the study. The goodness of fit values, including $\chi 2/df$, RMSEA, AGFI, and NFI, are essential for assessing how well the model fits the observed data. These values help researchers determine whether the model provides an acceptable representation of the relationships between the measured variables in the given population. The model values, including fit, indicate the level of fit achieved by the model, with specific criteria for perfect and acceptable fit. This table serves as a critical tool for evaluating the validity and effectiveness of the structural equation model in explaining the augmented reality immersion experiences of university students studying in science education (Salar et al., 2020).

Furthermore, the $\chi 2$/df ratio, RMSEA, AGFI, and NFI are commonly used goodness of fit indices in structural equation modeling (Salar et al., 2020). These indices provide valuable information about the overall fit of the model, with lower $\chi 2$/df and RMSEA values indicating better fit. The AGFI and NFI values also contribute to the assessment of model fit, with values above 0.95 considered as indicative of a good fit. The model values presented in the table further enhance the understanding of the model's performance, with specific criteria for perfect and acceptable fit. Overall, the "Table 6; Goodness of fit value range and model value" plays a crucial role in evaluating the structural equation model developed in the study, providing valuable insights into the augmented reality immersion experiences of university students studying in science education (Salar et al., 2020).

Variables			
Dependent variable Flow	Direct effect	Indirect effect	Total effects
Independent variable			
Emotional investment	–	0.334	0.334
Focus of attention	0.461	–	0.461
Presence	0.424*	0.148	0.572*
$R^2 = 0.714$			
Dependent variable Focus of attention	Direct effect	Indirect effect	Total effects
Independent variable			
Emotional investment	0.725*	–	0.725**
Presence	0.321*	–	0.321**
$R^2 = 0.952$			
Dependent variable Interest	Direct effect	Indirect effect	Total effects
Independent variable			
Usability	0.223*	–	0.223**
Emotional investment	0.755*	–	0.755**
$R^2 = 0.677$			

*$p < .05$

Table 7: Direct, indirect, and total effects

The Table 7: Direct, indirect, and total effects in the paper "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar et al., 2020, presents the direct, indirect, and total effects of the variables on the flow experiences of university students. The table provides a comprehensive overview of the relationships between the variables, including focus of attention, presence, emotional investment, usability, and interest, and their impact on the flow experiences of the students. The direct effects show the immediate influence of each variable on the flow experiences, while the indirect effects demonstrate the mediating role of certain variables in influencing the flow experiences. Additionally, the total effects represent the overall impact of each variable, considering both direct and indirect pathways, on the flow experiences of university students studying in science education (Salar, Arici, Caliklar, & Yilmaz, 2020).

The findings presented in the Table highlight the complex interplay of factors that contribute to the flow experiences of university students using augmented reality technology in science education. It is evident that variables such as focus of attention, presence, emotional investment, usability, and interest play significant roles in shaping the flow experiences of the students. Understanding these direct, indirect, and total effects is crucial for educators and researchers aiming to enhance the effectiveness of augmented reality immersion experiences in science education. By recognizing the specific contributions of each variable, educators can design interventions and instructional strategies that optimize the flow experiences of university students, ultimately improving their learning outcomes and engagement with AR technology (Salar, Arici, Caliklar, & Yilmaz, 2020).

Section 13: Recommendations

The following recommendations can be made, based on the findings of the study "A Model for Augmented Reality Immersion Experiences of University Students Studying in Science Education" by Salar, Arici, Caliklar, & Yilmaz, 2020:

- Since most pre-service teachers declared that they did not know anything about AR implementation, it is recommended that pre-service science teachers are given training on AR before starting work.
- As it was found that focus of attention and presence affect the flow, designers are advised to design AR in a way which easily enables the capture of attention and reality perception in order to generate flow in the participants.
- The sense of presence must be created to achieve flow experience in AR applications. To achieve this, user profiles should be considered and familiar environments, objects, concepts, and visuals employed.
- The findings indicate that presence and emotional investment have a direct effect on focus of attention. Therefore, in AR applications, realistic objects and environments should be used to enable students to feel the presence, enabling them to establish an emotional connection with the application and improving levels of attention.
- Individual emotional attachment should be sustained to aid focus of attention in AR applications. Applications should be developed based on real-life situations and individuals' previous life experiences.

- This study shows that usability has a direct effect on students' interest. Therefore, it is recommended that a simple, uncomplicated, and straightforward design should be used in order to increase interest.
- According to the results of the research, 99% of the students use smartphones. Therefore, AR applications compatible with smartphones should be further developed.
- The research indicates that students' use of technology in daily life and in an education context is already at a high level. Therefore, the use of AR applications should be encouraged in all classes.
- As the sample size for this study was limited, it is important that model studies are conducted with larger samples in future studies.
- In this study, students' opinions may have been influenced by the novelty effect of short-term involvement with technology that was new to them. In future studies, it is recommended that students use AR technology for longer periods in order to eliminate as far as possible the effects of innovation.
- In future studies, all variables used in the model should be examined through qualitative research, and the correlations investigated in depth.

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